

Four-Packet Polarization and the PSD Cross-Term Obstruction

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Abstract

The prime-shell route carries four packet coordinates whose signed reassembly is not determined by their separate energies. We isolate this interface in exact Hilbert-space form. The four-packet Gram is positive semidefinite, its coefficient energy is a Rayleigh quotient, and a four-point polarization identity reconstructs every cross-term from four phase-labelled energies. A trace bound gives a sharp universal envelope. We then give two rank-one packet fixtures with identical diagonal and trace data but target signed energies 16 and 0. Thus PSD and trace information alone cannot certify signed reassembly. The result is a structural obstruction and a precise specification of the missing polarized cross-correlation theorem; no arithmetic L^2 advance is claimed.

Claim level. PROVED_STRUCTURAL_L1 / FOUR_PACKET_CROSS_TERM_OBSTRUCTION.

1 Four packet interface

Let V_0, V_1, V_2, V_3 be vectors in a complex Hilbert space. We use the convention

$$\langle x, y \rangle = \sum_k \overline{x_k} y_k, \quad G_{j\ell} = \langle V_j, V_\ell \rangle. \quad (1)$$

The packet index is the four-sign interface inherited from the prime-shell lift. For a coefficient vector $c = (c_0, c_1, c_2, c_3)$, define the reassembled vector

$$V(c) = \sum_{j=0}^3 c_j V_j.$$

The desired signed estimate concerns $\|V(c)\|^2$, not just the four diagonal masses $\|V_j\|^2$.

Theorem 1 (PSD packet Gram). *The matrix G is Hermitian positive semidefinite and*

$$\|V(c)\|^2 = c^* G c \geq 0. \quad (2)$$

Proof. Hermitian symmetry follows from conjugate symmetry of the inner product. Expanding the norm of $V(c)$ gives (2), and positivity is immediate from the norm square. $\square$

The diagonal of G records the four individual packet energies, while the off-diagonal entries are the signed cross-terms. TPC-221's Schur result controls an absolute collision envelope, but does not determine these phases.

2 Exact four-point polarization

Theorem 2 (Four-point polarization). *For all vectors x, y ,*

$$\langle x, y \rangle = \frac{1}{4} \sum_{r=0}^3 i^{-r} \|x + i^r y\|^2. \quad (3)$$

Proof. Expand

$$\|x + i^r y\|^2 = \|x\|^2 + \|y\|^2 + i^r \langle x, y \rangle + i^{-r} \langle y, x \rangle.$$

Multiplication by i^{-r} and summation over $r = 0, 1, 2, 3$ kills the diagonal terms and the $\langle y, x \rangle$ term, because $\sum_r i^{-r} = \sum_r i^{-2r} = 0$. The remaining term is $4\langle x, y \rangle$. $\square$

Identity (3) is an exact compiler, but it carries four phase-labelled energies. Replacing those energies by one unsigned majorant discards precisely the information needed for the cross-term.

Corollary 3 (Trace envelope). *For every coefficient vector c ,*

$$0 \leq c^* G c \leq \text{tr}(G) \|c\|_2^2. \quad (4)$$

The constant is sharp.

Proof. The eigenvalues of a PSD matrix are nonnegative and their maximum is at most their sum. Apply the Rayleigh quotient bound to (2). A rank-one Gram with one nonzero eigenvalue attains equality. $\square$

3 Same trace, different signed reassembly

Theorem 4 (Scoped cross-term obstruction). *Let $u = (1, 0)$ and define two four-packet families by*

$$V_j^+ = u, \quad V_j^- = (-1)^j u, \quad 0 \leq j \leq 3.$$

Their Gram matrices are PSD and have the same diagonal $(1, 1, 1, 1)$ and trace 4. For $c = (1, 1, 1, 1)$, however,

$$\|V^+(c)\|^2 = 16, \quad \|V^-(c)\|^2 = 0. \quad (5)$$

Proof. Both families are rank one, so their Gram matrices are respectively J_4 and ss^* with $s = (1, -1, 1, -1)$. Each diagonal entry is one. The plus sum is $4u$, while the minus sum is zero, proving (5). Formula (3) holds exactly for every pair in both families, so the difference is genuine signed phase information rather than an algebraic inconsistency. $\square$

The plus fixture saturates (4): $\text{tr}(G)\|c\|^2 = 4 \cdot 4 = 16$. The minus fixture has the same trace but complete cancellation. Therefore no theorem whose inputs are only packet diagonals, trace, or an unsigned PSD envelope can identify the signed reassembly. This is a finite scoped obstruction; it is not an assertion that these two abstract phase patterns are both generated by every growing literal prime shell.

object	finite scope	status
packet Gram	4 coordinates, two rank-one fixtures	PSD exact
polarization	16 pair identities per fixture	exact
trace envelope	$\text{tr}(\mathbf{G}) = 4$	sharp
signed target	energies 16 and 0	non-identifiable

Table 1: TPC-222 certificate scope and claim boundary.

4 Certificate and route position

The exact certificate uses Gaussian-rational pairs, checks all sixteen polarization residuals for each fixture, verifies rank-one PSD Gram structure, and independently replays the trace and signed-energy contrast.

The strongest positive result is the exact four-energy polarization compiler. The strongest obstruction is same diagonal/trace with different signed cross-term behavior. The next theorem must control polarized literal packet energies or their signed cross-correlations.

ARITHMETIC_ADVANCE=NO, L2=NONE, FULL_GATE_B=OPEN.

5 Conclusion

TPC-222 closes the PSD-only shortcut at the four-packet interface. Polarization tells us exactly what data would recover a cross-term, while the paired rank-one fixtures show why trace and diagonal control are insufficient. A literal growing-scale signed cross-correlation estimate remains the open bridge to the prime shell.

References

- [1] TPC-218 repository proof record, “Prime-shell Hilbert lift and the sharp collapse barrier,” 2026. Internal source lock; no external arithmetic estimate is invoked.